SAFEMAN · STUDENT MODULE

SM-LOTO

Lockout / Tagout Energy Control

SafeMan™ by Oryx Institute · safeman.oryxinstitute.org

Theory + Isolation Practicum · Pass mark 70% · 2 modules

Name _____

Student no. _____ Class _____



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WELCOME

SafeMan Hazardous Energy Control (Lockout / Tagout)

SafeMan Hazardous Energy Control (Lockout / Tagout) — Theory + Isolation Practicum. This SafeMan course is engineered for corporate audit readiness against: GN 156 Machinery Regulations; NTA Unit Std 2061; ISO 45001 Cl. 8.1.

REGULATORY DISCLOSURE — READ FIRST

SafeMan™ is a proprietary inter-organizational safety management and operational competency benchmark developed by Oryx Institute (safeman.oryxinstitute.org). Courses are engineered in alignment with ISO 45001:2018, ISO 14001:2015, ISO 31000, ILO-OSH guidelines, and pre-referenced against Namibia Training Authority (NTA) Unit Standards and Namibian Government Notice 156 of 1997 for corporate audit readiness. SafeMan credentials verify industry competency and attendance between organizations and are non-NQA credit-bearing.

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How to use this module pack

Work through the modules in order. Each module states its focus and the key points you are expected to carry into practice. Diagrams referenced as web assets accompany the online reader at safeman.oryxinstitute.org. After the modules, sit the knowledge check: 10 questions, pass mark 70%, closed book.

This is a practicum course: after passing the knowledge check you must complete the assessor practicum checklist with a qualified assessor before your SafeMan credential is issued. A practical demonstration video slot is reserved in the online reader for every module.

MODULES

Module 1 — Energy sources & the 6-step isolation

All the ways a machine can move, and the disciplined sequence that makes it safe to touch.

Key points

Energy families: electrical, mechanical, hydraulic, pneumatic, thermal, chemical, gravitational — including stored energy in springs, accumulators and suspended loads.

6 steps: identify energy sources, notify, isolate, lock and tag, dissipate/ restrain stored energy, verify zero energy (try-out).

One worker, one lock, one key — personal locks only; a tag warns, a lock stops.

Group lockout box holds the keys; each worker applies a personal lock to the box.

Diagram: /assets/webp/sm-loto-padlock.webp

Practical video recording scheduled — follow the step-by-step illustrated checklist below.

Module 2 — Multi-lock, tags & try-out

Shared isolations under group control and the physical proof that zero energy is real.

Key points

Danger tags carry name, date, department, reason — and are never removed except by their owner per procedure.

Group boxes and hasps let many workers share one isolation with individual locks.

Try-out: attempt start controls, test with instruments, check bleeds — prove it dead before skin goes in.

Restoration is the mirror image: tools out, guards on, people clear, locks off in reverse order, notify.

Diagram: /assets/webp/sm-loto-padlock.webp

Practical video recording scheduled — follow the step-by-step illustrated checklist below.

ASSESSMENT

Knowledge check

The closed-book knowledge check draws 10 questions from the bank reproduced in Appendix A. Pass mark 70%. Your SafeMan credential is issued after the knowledge check and the practicum sign-off, and remains valid for 24 months from issue.

Assessor Practicum Checklist — Lockout / Tagout

Pass standard: Isolation executed end-to-end on the training panel to assessor standard.

1	All isolation points identified from the provided P&ID / switchgear diagram.	
2	Multi-hasp and personal lock applied with a completed danger tag (name, date, reason).	
3	Stored energy dissipated: bleed, discharge, block or lower as applicable.	
4	Physical zero-energy try-out performed on the start controls.	
5	Restoration sequence demonstrated in correct reverse order.	

Assessor name, role/qualification and company are captured on the sign-off screen of the SafeMan platform and bound into the QR-verified certificate record.

APPENDIX A — QUESTION BANK

Knowledge check bank

Q1. The final LOTO step before work is:

A. Tagging

B. Verifying zero energy by try-out

C. Telling the supervisor

D. Signing the log

Q2. Stored energy types include:

A. Only electricity

B. Springs, accumulators, suspended loads, residual pressure, gravity

C. Only hydraulics

D. None — isolation removes all

Q3. 'One worker, one lock, one key' means:

A. Share keys in the team

B. Each worker applies a personal lock with a unique key

C. The foreman holds all locks

D. Tags replace locks

Q4. A danger tag's function is to:

A. Stop the machine

B. Warn and identify — the lock does the stopping

C. Decorate the panel

D. Record wages

Q5. Group lockout boxes exist so that:

A. Keys are not lost

B. Every worker under a shared isolation holds a personal lock on the box

C. One lock suffices

D. Contractors skip locks

Q6. Try-out includes:

A. Visual check only

B. Start-button attempt plus instrument verification and bleed checks

C. Asking the operator

D. Waiting 5 minutes

Q7. Locks are removed:

A. By the shift boss on request

B. Only by their owner through procedure

C. At shift end by anyone

D. By the client

Q8. Which energy is often forgotten?

A. Electrical

B. Gravitational — suspended or raised loads

C. Steam

D. All are remembered

Q9. Restoration sequence is:

A. Reverse of isolation after guards on and people clear

B. Random

C. Fastest path

D. By the newest worker

Q10. Isolation points are identified from:

A. Memory

B. P&IDs / switchgear schedules during planning

C. Guesswork

D. The permit's mood

Answer key

Q1=B Q2=B Q3=B Q4=B Q5=B Q6=B
Q7=B Q8=B Q9=A Q10=B